MUHAMMAD GEMILANG RAMADHAN

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Summary

Electrical Engineering student with a strong interest in autonomous systems, path planning, applied robotics, and software development, particularly through thesis work on USV autonomous navigation using Lazy Theta Star and SAPF-DDPG. Experienced in algorithmic problem-solving, simulation-based analysis, and technical implementation, with strong communication skills and the ability to present complex concepts clearly.

Education

SEPULUH NOPEMBER INSTITUTE OF TECHNOLOGY | Surabaya, Indonesia | Graduating in September 2026
Electrical Engineering, Control System Specialized

XAVERIUS 1 PALEMBANG SENIOR HIGH SCHOOL | Palembang, Indonesia
Science-focused study

Skills

- **Front-End & Back-End Development:** React, JavaScript, Tailwind, Bootstrap, Vite, Node.js, Express
- **Programming & Data:** Python, SQL
- **Machine Learning:** TensorFlow, PyTorch, OpenCV
- **Microsoft:** Excel, Word
- **Tools & Workflow:** Git, GitHub, Jira, Notion
- **Languages:** Indonesian (native), English (fluent)

Work Experience

Kulkas Babe | Multi-Tenant E-Commerce Front-End | June 2025 - August 2025

(freelance project)

- Sole front-end in a 2-person team to build responsive, dynamic shopping UI/UX with Laravel Blade, Alpine.js, Tailwind, vanilla JS.
- Designed payment flows by integrating Midtrans API (QRIS), Google Maps API (distance/fees).
- Partnered with back-end to enforce secure, maintainable components, versioned patterns, and clean code reviews.

Anargya ITS EV Team | Web Developer & Designer | January 2024 - February 2024

- Shipped the team's public website; translated Figma designs into responsive, accessible pages using Bootstrap.
- Built a lightweight brand system (colors, type scale, spacing) to keep web and Instagram assets consistent.
- Produced social assets (feed/story carousels, sponsor spotlights) that supported announcements and partner visibility.

Projects

USV Path Planning using Lazy Theta Star and SAPF-DDPG | Undergraduate Thesis | January 2026 - July 2026

- Developed a USV path planning system by combining Lazy Theta Star for global path planning and SAPF-DDPG for adaptive local obstacle avoidance.
- Implemented simulation-based navigation using a 4-DOF USV model, static and dynamic obstacle scenarios, and G2 Continuous Bézier Spiral smoothing for smoother trajectory generation.
- Trained DDPG to tune SAPF parameters such as attractive gain, repulsive gain, and safe distance to improve obstacle avoidance behavior in dynamic environments.
- Evaluated the system through simulation and field implementation on the USV LSS-01 platform at Kolam InfinITS ITS.
- **Stack:** Python, NumPy, Matplotlib, OpenCV, reinforcement learning, path planning, USV control.

JejakBatik | Machine Learning Pattern Recognition | December 2024
(Bangkit Capstone project, team of 5)

- Architected the end-to-end ML pipeline (data prep, augmentation, training, evaluation) and designed an EfficientNetB0 classifier, achieving 85% pattern-recognition accuracy.
- Implemented real-time image processing & classification with TensorFlow (and TensorFlow JS for mobile preparation) + OpenCV, integrating the model with the Android app for on-device inference.
- Tuned preprocessing and caching to keep inference responsive.
- **Stack:** Python, TensorFlow (EfficientNetB0), OpenCV.

Certification

- **Bangkit Machine Learning Path Graduate | 2024**
  Completed intensive training in machine learning, TensorFlow, supervised and unsupervised learning, OpenCV-based image processing, and pattern recognition.
- **Udemy 100 Days of Python | 2023**
  Completed Python programming course covering fundamentals, functions, automation, and project-based development.
- **MySkill Microsoft Excel | 2023**
  Completed training in pivot tables, Power Pivot, and basic data visualization.
- **MySkill SQL for Data Analysis | 2023**
  Completed training in SQL fundamentals, data querying, table manipulation, and basic analysis.